

Table of Contents

1. General interleaving rules
2. Interleaving techniques table
3. Technique explanations
4. Understanding mastery levels
5. Understanding knowledge types

General interleaving rules

Interleaving your retrieval is one of the most important parts of performing retrieval. Retrieval without appropriate interleaving can in some cases void the effectiveness of the retrieval session. What that means is that non-interleaved retrieval that does not produce sufficient cognitive load can be practically useless. When doing your interleaved retrieval:

- Minimise the usage of techniques aimed at low levels of mastery. You will be able to fulfil these lower levels when doing higher levels anyway, whereas low levels do not take you to higher levels.
- Apply general rules for appropriate spacing. Good quality interleaving is more important than accurate spacing. Your schedule can be off by days or even weeks, as long as the quality of each session is good.
- Better retrieval techniques generally involve higher cognitive load and test on multiple relationships with a focus on big picture integration, justification and prioritisation of ideas, and back-and-forth thinking between multiple concepts. Any technique that ticks these boxes (even if not listed here) is likely to be appropriate.

- Use your learning objectives from your curriculum (if you have one) to gauge how “out of scope” to do your retrieval. Aim to go slightly out of scope for optimal results as staying too rigidly in the scope of your curriculum can make it harder to re-encode complex concepts.

Interleaving techniques table

Search or sort the table based on your desired subject type or knowledge mastery level to see the appropriate techniques. Use the contents menu above or scroll down to read about each technique, the different levels of mastery, or different knowledge types.

Search

Knowledge type	Level of mastery	Example techniques
Declarative	0. Waste of time	Passive re-reading and re-writing, passive re-listening, practice questions (passive method), peer/group discussion (undirected)
Declarative	1. Low levels	3C's, flashcards (simple), brain dump (linear), generated questions (isolated), teaching (isolated), Feynman technique, MoL or mMoL, story/link method, Ben system
Declarative	2. Mid levels	Flashcards (simple relational), brain dump (mindmap), generated questions (simple relational or multi-relational), teaching (simple relational), practice questions (direct method)
Declarative	3. High levels	Chunkmaps, generated questions (evaluative), modified WPW, peer/group discussion (evaluative), practice questions (extended method)
Procedural	0. Waste of time	Re-reading and re-writing notes, REBIM

Knowledge type	Level of mastery	Example techniques
Procedural	1. Low levels	Retrieved execution (simple), simple flashcards, Ben system, challenges (simple)
Procedural	2. Mid levels	Retrieved execution (integrative), challenges (integrative), variable modification
Procedural	3. High levels	Retrieved execution (applied), challenges (edge-case), modified WPW, variable addition

Technique explanations

Technique	Description
REBIM (Repetitive Execution Beyond Initial Mastery)	A technique coined by Dr Justin Sung, REBIM refers to the repetitive completion of tasks or activities for skills and processes that can already be done to a high level of proficiency. For example, coding the same simple scripts repeatedly or solving the same difficulty of maths question, or engaging in the same kind of dialogue or conversation for languages. The benefit of the technique is negligible considering the alternative techniques that could be used instead.
Passive re-reading and re-writing or passive listening	Simply reading and re-writing notes without any particular cognitive focus. Usually boring, monotonous, disengaged, and repetitive. The focus may be on trying to just memorise what is being written, repeatedly saying the sentences mentally, or just "trying to understand". This does not activate most effective learning pathways and is highly time-consuming and demotivating.
Practice questions (passive method)	Solving problems and answering questions and immediately checking the answer. Often paired with answering the question mentally only, or just concluding "I could solve that" without going through all of the steps, then checking the answers to see if it makes sense. The illusion of learning is that because the question seems familiar and the answer makes

Technique	Description
	sense, that you are able to solve it in a realistic retrieval setting, however this is not true due to the gap between retrieval competency and perception of competency (the illusion of explanatory depth). A common way of using practice questions, but limited effectiveness compared to alternatives.
Practice questions (direct method)	Solving problems and answering questions and checking the answers immediately after the attempts. If incorrect, content material may be focally reviewed, but more often if it "makes sense" then it is concluded that it was just a silly mistake. The actual attempt at answering activates generation cognitive effects, however when the true cause for mistakes is not properly addressed, we can run into "silly mistake syndrome" whereby we continue to make mistakes that cost valuable marks, because the reason for the mistakes remains unaddressed. Students can feel frustrated at these silly errors that occur regularly.
Practice questions (extended method)	After answering practice questions to your best attempt, an answer sheet is created by reviewing any answers you feel unsure about and re-studying the relevant content. Your answer sheet is then checked against the official answer sheet. This creates generative effects at two levels, while also creating targeted revision points on weaker topics. This can easily be done as a group activity as well when practice questions are not available, with each member creating a set of practice questions and answer sheets for other members. This is one of the best ways to use practice questions with incomparably higher levels of learning compared to passive and direct methods.
3C's (Cover, copy, check)	A low-level "fill-in-the-blanks" style retrieval technique whereby you cover part of your notes, then attempt to rewrite or redraw the covered part from memory. You then uncover the notes and check. This has been shown to be effective at improving short-term retention on basic information, especially among younger children. There is insufficient theoretical or empirical data to suggest this is an effective or time-efficient technique at higher levels.
Flashcards (simple)	Flashcards, either physical or through a digital medium such as Anki (preferred). The characteristic feature of simple flashcards is that each card tests on isolated facts, concepts or processes. Concepts do not relate to each other. Often useful for specific definitions, constants or other highly specific pieces of information that do not fit into relational notes. When other aspects of the learning system are inadequate, there is often an overload of simple flashcards causing enormous time-waste through repetition and steep forgetting curves.

Technique	Description
Flashcards (simple relational)	Flashcards, preferably digital through a medium such as Anki, whereby each card tests on the relationship or dynamic or influence that two or three concepts have with each other. Useful for key relationships that are highly specific and difficult to remember. These are often processes that are conceptually separate from the rest of the topic and are therefore hard to represent through relational note-taking as they can not fit into clean chunks. Relatively rare use-cases. This is also the highest level of flashcards usage we recommend as more complex forms of knowledge mastery are best retrieved through other techniques.
Brain dump (linear)	Type out everything you know about a certain topic. It's recommended that you follow learning objectives as a guide on what to brain dump on. Brain dumps can be on individual concepts or entire topics, depending on available time. Typing is recommended for speed with this technique. An easy but surprisingly useful technique for testing on mass retrieval at low to low-mid levels of knowledge mastery. It is limited in being able to accurately describe multiple relationships due to the inherent limitation of linear notes. It is also inherently time-consuming as gap identification relies on "dumping" until we stumble upon an area we have difficulties retrieving. Recommended as an end-stage retrieval form as alternatives can provide faster gap identification.
Brain dump (mindmap)	The same as a linear brain dump, but in a mindmap form. This is not as good at covering the smaller details, but much better at covering relationships and chunks. As more difficult assessment and retrieval standards force you to think in relationships and chunks, this form of brain dumping is better to identify high-yield gaps to fill before moving into more detailed forms of retrieval. Mindmap brain dumps are a good early to mid-stage retrieval technique.
Generated questions (isolated)	Creating your own practice questions. Isolated questions are those that focus on retrieving single facts, concepts and explanations of isolated processes. Many "what" based questions tend to be isolated. These questions are not very worthwhile as alternatives produce the same level of detailed learning and more.
Generated questions (simple relational or multi-relational)	Creating your own practice questions. Simple relational questions challenge you to explain or discuss the relationship between two related ideas, whereas multi-relational questions challenge you to retrieve the relationships between more than two related ideas.

Technique	Description
Generated questions (evaluative)	Creating your own practice questions. Evaluative questions challenge you to not only explain and discuss relationships between multiple related ideas, but also challenge you to compare and contrast the strength and importance of the relationships, making judgements and prioritising different concepts against each other. The key difference between evaluative and multi-relational is that the relationships in multi-relational are acknowledged and explained, but judgements on importance are not made. This is the highest form of questioning and there is significant cognitive load and generation effects in the act of both creating the question and answering the question. It is recommended to have this form of interleaved retrieval in early-stage revision schedules so that answering the questions can be included as a mid to late stage retrieval option. This creates time-efficiency and when answering the questions, it can be combined with the practice questions (extended method) technique.
Teaching (isolated)	Teaching the technique to a friend or imaginary student. Teaching to an imaginary student is preferred because when teaching to a friend, they may fill in gaps for you with their own knowledge, instead of forcing you to realise that you are unable to explain something in a full level of depth. In the isolated version of teaching, explanations are limited to isolated facts, concepts and processes.
Teaching (simple relational)	The same as the isolated version, however instead of explanations being limited to isolated facts, concepts and processes, we are now explaining and discussing how different concepts and processes relate to each other. This is the highest form of teaching retrieval that can normally be reached for learners that do not have a high-intermediate level of encoding, especially with regards to the layers of learning taught in the late Ascent stages. This is because the more advanced version (WPW reteaching) is extremely challenging to perform correctly and is nearly impossible without ultra-refined chunking ability. Simple relational teaching is a relatively effective form of retrieval and should be incorporated in the early to mid stages of revision scheduling as it can find gaps in knowledge faster than many other forms of retrieval.
WPW reteaching	Taught in late stages of the course. An advanced retrieval method that is essentially impossible for any students without near mastery beyond Ascent 3 level. It challenges on all levels of learning with rapid gap identification. This is ideally an early to mid stage retrieval form due to how efficiently it identifies gaps in knowledge. Warning: attempting this technique without mastering earlier stages will result in virtually guaranteed technique failure. It will certainly be more time-consuming and less efficient than simple relational

Technique	Description
	teaching, until your skill level increases naturally to the appropriate stage. WPW reteaching is one of the most challenging techniques taught in the course, with considerable difficulty even for those who have mastered all earlier stage techniques. Those who have the prerequisites to perform WPW reteaching correctly are likely already in or near the top 1% of academic achievement region or nationwide. In short, do NOT rush the course just to learn this technique.
Feynman technique, method of loci (MoL) or modified method of loci (mMoL), link or story method and Ben system	All different memorisation techniques. Some of these are taught in the course, however many are not. The ones that are taught in the course are taught in late stages. Although these memorisation techniques are powerful for rote learning of isolated information, the use-cases are very narrow and there is a tendency to over-rely on these methods instead of developing proper encoding techniques which are several-fold more efficient at achieving higher knowledge mastery levels. None of these techniques facilitates learning at mid or high levels of knowledge mastery. They are a great trick to use once in a while but are heavily over-rated and very limited in real-world practical usefulness.
Chunkmaps (synonymous with GRINDEmaps)	Mindmaps created with correct chunking technique. These are normally a core part of the encoding process in our Bear Hunter System taught in the growth phase, however, if you have previously learned material before learning how to do correct encoding, it is worthwhile to re-encode all of your material into a chunkmap form to stimulate higher levels of mastery and improve retention. In this situation, this technique should be the very first step of the revision process and should be performed as early as possible, making time for other forms of interleaved retrieval. Most people find that they struggle considerably at creating a chunkmap with previously poorly encoded information, largely because it was originally never processed at higher levels. This tends to be reflected in poor overall time efficiency and undesirable results.
Peer/group discussion (undirected)	Most group study is a colossal waste of time compared to what could be achieved in the same time through alternative techniques. Although group study psychology does create some interesting cognitive benefits, we can achieve the same benefits and more with more directed methods.
Peer/group discussion (evaluative)	Group discussion whereby the goal is to critique, judge and argue the importance of different relationships and concepts, especially in view of the bigger picture. Methods of facilitating group study to achieve this are taught in the course.

Technique	Description
Retrieved execution (simple)	Retrieved execution is practising a skill or process from memory without any cue or guidance. The simple version is limited to isolated applications of the process without variation. It is often the first way in which practice a new skill, however many people will continue with this simple version beyond it's beneficial time frame.
Retrieved execution (integrative)	Retrieved execution is practising a skill or process from memory without any cue or guidance. The integrative version involves taking multiple skills and processes and bringing them together to achieve a greater or more complex function. This may be easy coding or languages where more complex functions can be coded using simpler, component skills, or common phrases and dialogues can be chained together to form a simple, but varied conversation. Learners should aim to move to integrative retrieved execution at the soonest possible time, immediately after gaining basic competency with simple execution. This provides stronger generative effects, greater cognitive load and more interleaving.
Retrieved execution (applied)	The applied version of retrieved execution is fundamentally based around a bigger picture, more complex application of the simpler skills and processes. While integrative practice brings together individual components to form a more complex product, applied practice starts with a wider project that requires dozens or hundreds of integrated components to make it function. In coding, this could mean developing a web application to fulfil a real need, requiring you to not only design and strategise how the functionality can be developed and executing on it, but also bug-fixing and continually evaluating the quality and efficiency of your work. For languages, this could be thinking of a common scenario in which you will require your language and practising multiple variations of conversations that could arise in that setting. For example, a simple meeting with friends at a restaurant may involves hundreds of variations of phrases to navigate the entire evening smoothly in your new language. Applied retrieved execution is the most challenging but most effective form of practice. Learners should aim to move towards applied execution once some level of competency is achieved with integrative execution. This transition from integrative to applied does not need to be rushed, as deficiencies in integrative practice can potentially cause quite significant efficiency consequences when attempting under fully applied execution challenges.
Challenges (simple)	Challenges are similar to retrieved execution, however are different in that challenges involve a prompt or problem stem that must be solved using your skills. Challenges can be self-created or externally provided. Simple challenges help to build basic procedural competency with single, isolated pieces of information. For example, coding a certain

Technique	Description
	function, mastering a line of syntax and a few variations, or introducing yourself and ordering a glass of water in a new language.
Challenges (integrative)	Challenges are similar to retrieved execution, however are different in that challenges involve a prompt or problem stem that must be solved using your skills. Challenges can be self-created or externally provided. Integrative challenges, like integrative retrieved execution, involve bringing together multiple component pieces to solve a more complex problem. These challenges sometimes involve using two or more component processes or skills in ways that were previously not directly taught. A common example is in solving physics or maths problems where different formulas and concepts come together to calculate a multi-step problem.
Challenges (edge-case)	Challenges are similar to retrieved execution, however are different in that challenges involve a prompt or problem stem that must be solved using your skills. Challenges can be self-created or externally provided. In edge-case challenges, the problems are centered around atypical examples of how the skills and processes can be used. They are almost always different from how the skills were originally taught and the learner needs to think about how certain processes and skills apply in this situation. Strong declarative knowledge on the principles is often needed to navigate this. The ability to logically reason that a certain approach is viable and makes sense is necessary to even make a start. For maths or physics, these may be challenge questions that involve multiple concepts being applied in non-textbook ways. For coding, this could be in creating advanced functionality by using a series of simple functions, rather than resorting to more powerful functions (that you may not know yet). For languages, this may be conversations or reading/writing in contexts that are unfamiliar or more advanced, such as business news, political debates, controversial opinions, or atypical responses to common questions. Edge-case practice is primarily useful for the most challenging kinds of retrieval. This may be unnecessary for your learning needs.
Variable modification	Variable modification simply means changing the variables in a given problem, challenge or question so that you can solve the same problem again, with different figures. The nature of the problem is the same and approach is the same. This is good for practising low-level procedural skills before moving into more integrative stages of retrieval.
Variable addition	Variable addition involves adding variables to an existing problem, challenge or question to increase it's difficulty. The variable added needs to make logical sense (a process which involves cognitive load and positive effects on learning), while also being potentially

Technique	Description
	solvable. Strong generation effects are activated at multiple points of this technique, including identifying which variable to add to make the problem more complicated, whether the variable makes sense, what value to give the variable in order for it to be reasonable, whether a solution actually exists when this variable is added, and evaluating how the variable interacts with other existing variables. More than one variable can be added if it is viable. This technique is most obvious for problems like in maths, physics, accounting, economics, finance, business, etc. but it is also applicable for coding and languages. For example, in coding, there may be a challenge to create a pop-up window when a query results in a positive value. A variable could be added so that if no query is delivered within a time-out period, a different message appears. This can be further complicated by adding an input variable where the query result changes depending on a user input. In languages, we can add variables such as number of speakers in a conversation, number of view points of an opinion, level of detail and conditions within statements or questions during conversation (e.g. "where do you want to eat" vs "if you're hungry, we can eat here, otherwise you can pick somewhere else" vs "I've already eaten here before, but I don't mind eating here again. If you're hungry, we can eat here, otherwise you can pick somewhere else, as long as it tastes good.")

Understanding mastery levels

Overview

Different levels of mastery correspond to different strengths of encoding, retention and ability to retrieve. Lower levels tend to perform more poorly overall while higher levels are deeper, retained longer and more nuanced in retrieval forms. Low levels of mastery are sometimes necessary when the type of knowledge demands it, but a common misconception is that low levels of mastery need to be focused on before moving to higher levels. This is not the case as our brain is able to automatically fulfil low levels of mastery on the way to high levels.

For example, in attempting to solve a complex problem, our brain will automatically remember the equation we used, as well as the definition of a concept. This is usually faster than if we tried to focus on just low levels, to begin with, because of how human cognitive architecture functions. This is

because retention of information depends on whether the information belongs in structure and has relevance (among other factors). Because techniques that exclusively work on low levels do not provide structure and relevance, it takes longer to remember information than if a higher-level technique were used to achieve the same effect.

Critically, techniques that are suited for low levels are ineffective at helping learners reach higher levels. If you are assessed at higher levels, your retrieval should also be at equivalent levels. We don't expect someone to do a million flashcards and suddenly write a PhD.

Low levels of mastery

These levels are characterised by isolated information without relationships (or very superficial relationships). Many curriculums for primary age students assess only low levels of knowledge mastery. During secondary and tertiary education, low levels of mastery become ineffective at gaining top marks and inefficient due to the disproportionate increase in total volume of facts to be learned, relative to working memory capacity. Higher order learning techniques can increase the efficiency of knowledge encoding, which solves this problem at the root.

Examples of low levels of mastery:

- Simple recall of facts and figures
- Describing or listing things
- Explaining single concepts and processes

Middle levels of mastery

In these levels, we are starting to relate ideas together and seeing how concepts influence each other. Concurrently, knowledge is now being applied to contexts outside of how it was immediately learned.

Examples of middle levels of mastery:

- Basic problem solving
- Discussing and explaining the relationship concepts have on each other
- Explaining or demonstrating how different components influence each other

High levels of mastery

At high levels of mastery, we are not only relating ideas, but we are connecting the relationships with the big picture of the topic. We are evaluating ideas and making judgements about the relative importance of different relationships and concepts. We are not just aware that relationships exist, but we are able to prioritise these relationships, manipulate the information, group and chunk concepts together and make judgements about the underlying logic of the topic.

Examples of high levels of mastery:

- Comparing and contrasting the relative advantages and disadvantages of one concept compared to the alternatives, and how this changes depending on the context
- Discussing the impact that changing one variable can have in the variables of other related concepts
- Discussing how one concept can influence the overall function of a larger mechanism, and how this differs to the influence that other concepts have
- Solving problems that involve utilising multiple concepts together in novel combinations

Understanding knowledge types

Declarative

Declarative knowledge is information that is based on facts, concepts and logic. It is the type of knowledge that involves encoding content material. Declarative knowledge is sometimes also called conceptual knowledge, or descriptive. One of the characteristics of declarative knowledge is that the information itself remains static. For example, if you learn about how a kidney functions, the nature of this information does not change, since the facts are already set. Your understanding and nuanced appreciation of the information may change, the facts remain static. This is unlike procedural knowledge whereby the knowledge on how to execute on certain processes innately changes as mastery grows.

Examples of predominantly declarative knowledge:

- Science subjects
- History
- Politics
- Law
- Economics theory
- Accounting theory

- Mathematics theory
- Business and finance theory
- Coding logic
- Language vocabulary, syntax, grammar and structure

Procedural

Procedural knowledge is stored in different parts of the brain than declarative knowledge and involves some related processes, as well as some different cognitive processes. Procedural knowledge is based on actions, behaviours and processes, rather than facts or concepts. It is the knowledge on how to perform skills or procedures. In most settings, both declarative and procedural knowledge are required, but different subjects and settings require different levels of each.

Examples of predominantly procedural knowledge:

- Conducting a lab experiment with technical precision (the theoretical knowledge on how the experiment should be conducted is declarative, while the skill of conducting it accurately and precisely may be procedural)
- Driving or riding a bike
- Cooking or baking (the theoretical knowledge of how to cook or bake a meal is declarative, while the skill of executing on the theory is procedural)
- Study skills (the theoretical knowledge of learning principles and how techniques should be performed is declarative, while the skill of actually performing them in practice is procedural)
- Solving mathematics or physics problems (the theoretical understanding of how to approach a problem and how different concepts can be brought together to create a solution is declarative, while the ability to actually work out the answer through calculation is procedural)
- Speaking a language fluently or writing fluently (the vocabulary, grammar and syntax knowledge is declarative, while the skill of speaking or writing is procedural)
- Creating an accounting balance sheet accurately and efficiently
- Building an app or developing something with coding languages (the theoretical understanding of the language logic and syntax is declarative, while the skill in executing on it accurately and efficiently is procedural)

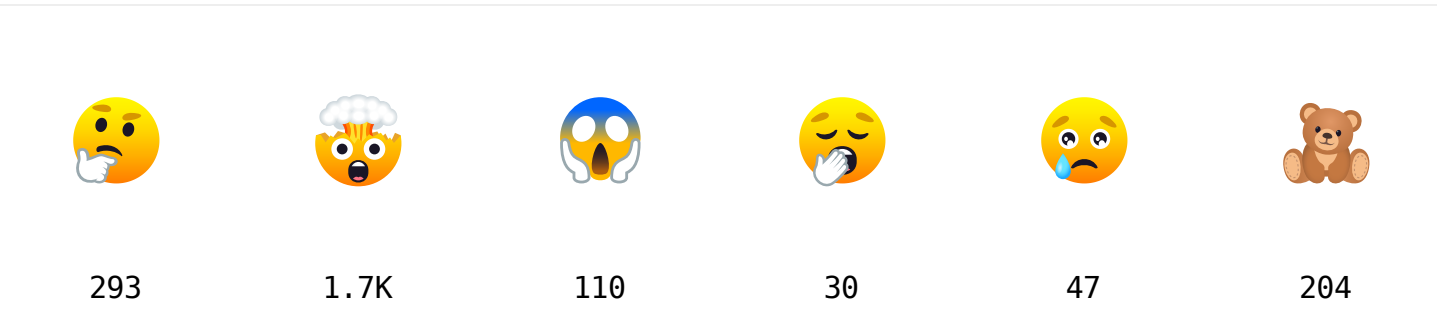
As you can see, most applications of knowledge require both sound declarative and procedural knowledge. If someone has great procedural knowledge of mathematics, but poor declarative knowledge of the concepts, they may be able to solve problems easily when they are familiar, but they will struggle to approach new types of challenges or understand how different mathematics concepts can be utilised. Likewise, someone

with great declarative knowledge without procedural knowledge may know how to approach a problem but may struggle to properly calculate an answer.

In another example, someone with great procedural coding knowledge may be able to easily write lines of script for simple functions that they have practised many times, but they may struggle with coding challenges, technical interviews, and developing new projects outside of their familiar scope. On the other hand, if this person has high declarative knowledge and poor procedural knowledge, they may know how to build a new and innovative function or project correctly, but will struggle with invalid lines of code, bugs, inefficient coding practices and poor total coding productivity.

Clearly, in both examples, a truly excellent learner demonstrates both.

How was that?



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